

MortgageOS™ v8.0.0 Architecture Blueprint

Author: Technical Engineering Architecture Lead
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Framework: XRPL Native MPT + Hooks Integration

1. Core Paradigm Shift: Native Multi-Purpose Tokens (MPT)

v8 deprecates standard NFTs (NFTokenMint) which function as opaque handles. Instead, it transitions completely to the XRPL native MPT (Multi-Purpose Token) architecture. This shift allows the system to establish asset-backed debt and balance mechanics natively on-ledger, supporting built-in compliance freezing, clawback rules, and cross-account balance metrics without maintaining massive, high-risk off-chain tracking databases.

2. Dynamic Escrow & Variable Tax Impound Handling

Amortizing real estate property tax metrics over 30 years introduces catastrophic bookkeeping risk due to volatile municipal rate and valuation changes. v8 implements an isolated annual review cycle conforming to **RESPA Section 10 (12 CFR 1024.17)** rules. The tax engine uses a 12-month projected window, calculating 1/12th collections based on the previous year's actual assessment data. If shortfalls occur mid-year, the servicer advances the funds immediately to remain compliant, recovering the delta through secondary pass-through billing modifications.

State Variable Key	Data Type	Functional Responsibility
TAX_DUE_DATE	Unix Timestamp	Enforces strict date-locking; blocks early capital drainage.
TAX_AUTHORITY_ID	20-Byte Account ID	Explicitly designates the authorized recipient address.
INITIAL_TAX_RATE	Basis Points (BPS)	Initial baseline tax percentage setting.
LAST_VALUATION	Exact Cents (Int)	Baseline home value mapping (Defaults to purchase price).

3. Decentralized Metadata Infrastructure (IPFS/Filecoin)

To guarantee that your legal files remain as permanent and immutable as the ledger records, v8 introduces an automated pipeline using Python to pin complete metadata manifests directly to IPFS and Filecoin nodes. The resulting CID hash is permanently stamped onto the asset token's core data profile, eliminating single points of web server failure.

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